BLOCKCHAIN

Hash Functions Visual Guide

SHA, Merkle trees, password hashing, and collisions — fingerprints for data.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

What Makes a Hash Function

Arbitrary input in, fixed-size fingerprint out — with four promises.

Deterministic	same input -> same output, every time
Fixed size	SHA-256: always 256 bits, input any length
Avalanche	flip one input bit -> ~half the output bits flip
One-way	infeasible to invert or find collisions (for crypto hashes)

STEP 2/6

The SHA Family

NIST standards, growing digest sizes — and retirements.

SHA-256

256-bit — Bitcoin, TLS, Git

SHA-512

512-bit — bigger digest, 64-bit optimized

SHA-3

sponge construction — different design, backup standard

SHA-1 / MD5

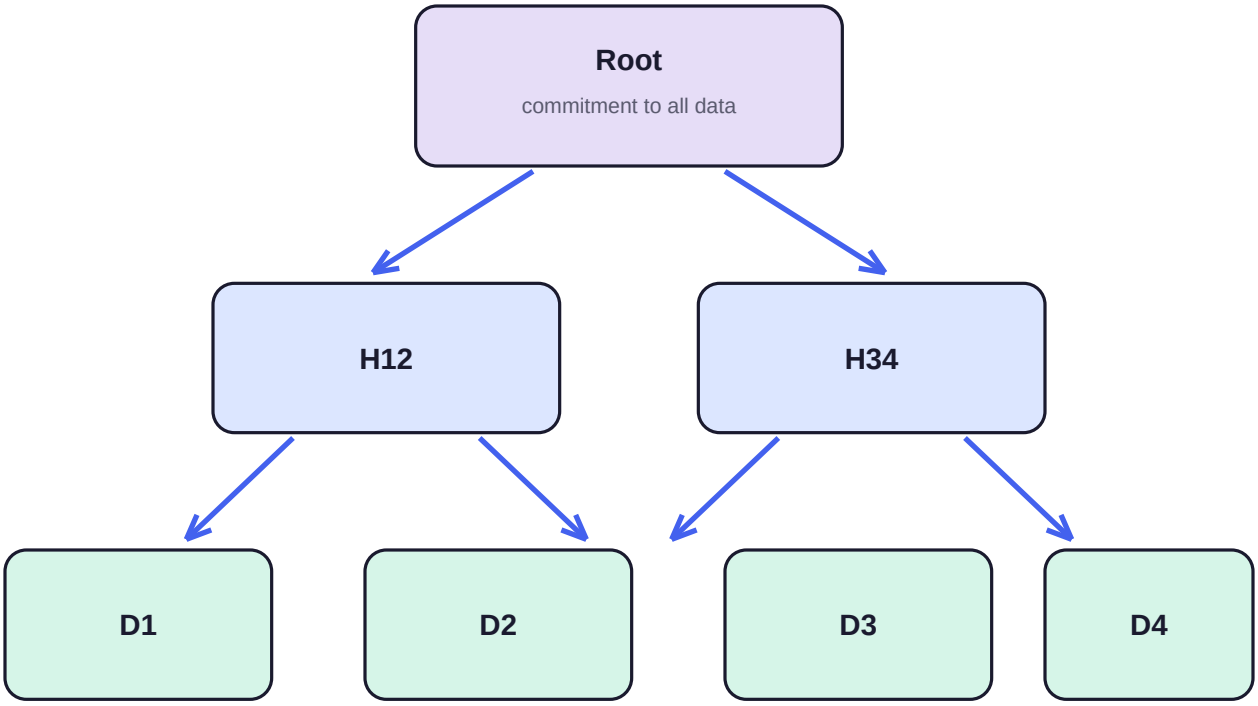
BROKEN — collisions demonstrated, do not use

Broken != useless everywhere: MD5 still fine for non-adversarial checksums — but never for security.

STEP 3/6

Merkle Trees

Pairwise hashing up to one root — the data structure of verifiable lists.



Prove D2 is in the set: reveal H(D1), H34, and the root — verifier recomputes.

Git commits, blockchains, certificate transparency: all Merkle trees underneath.

STEP 4/6

Password Hashing: Slow on Purpose

Fast hashes let attackers try billions of guesses per second. So we go slow.

SHA-256(password)

GPU: billions/sec — cracked in hours

bcrypt / Argon2

tunable work factor — thousands/sec by design

SALT

unique random per password — defeats rainbow tables

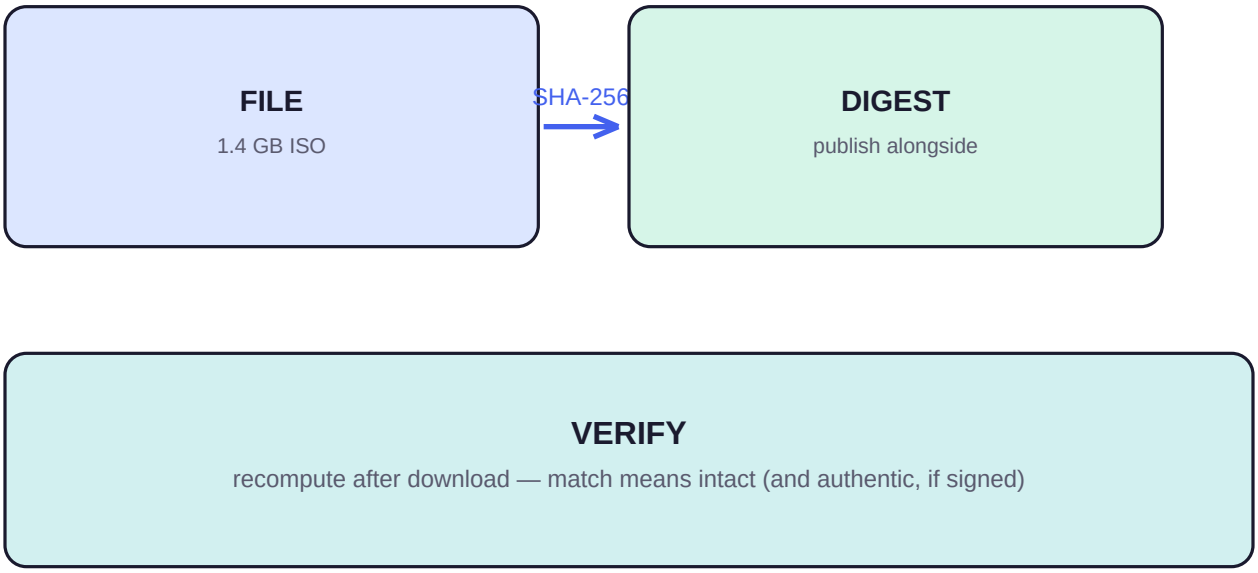
PEPPER (optional)

server-side secret mixed in — database leak alone is not enough

STEP 5/6

Data-Integrity Hashing: Fast on Purpose

Detect corruption and verify downloads — speed is the point here.



Same primitive, opposite goal from passwords: here fast verification is a feature, not a bug.

STEP 6/6

Collisions & the Birthday Bound

With n-bit hashes, collisions appear after ~2^(n/2) tries — not 2^n.

BIRTHDAY PARADOX

23 people -> 50% chance of a shared birthday (365 days)

SHA-256

collision at ~2^128 — still infeasible

MD5 (128-bit)

collision at ~2^64 — demonstrated in practice

Digest size halves under birthday attacks: 256-bit hashes give 128-bit collision resistance.

STEP ✓/✓

Key Takeaways

Hash Functions — Remember

- Crypto hashes promise determinism, fixed size, avalanche, and one-wayness.
- SHA-256/512 and SHA-3 are current standards; SHA-1 and MD5 are broken for security use.
- Merkle trees commit to whole datasets with one root and log(n) inclusion proofs.
- Password hashing must be SLOW (bcrypt/Argon2) and salted; fast hashes get brute-forced.
- Integrity hashing must be FAST: verify downloads and detect corruption cheaply.
- Birthday attacks halve effective strength: n-bit digests give ~n/2 bits of collision resistance.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev